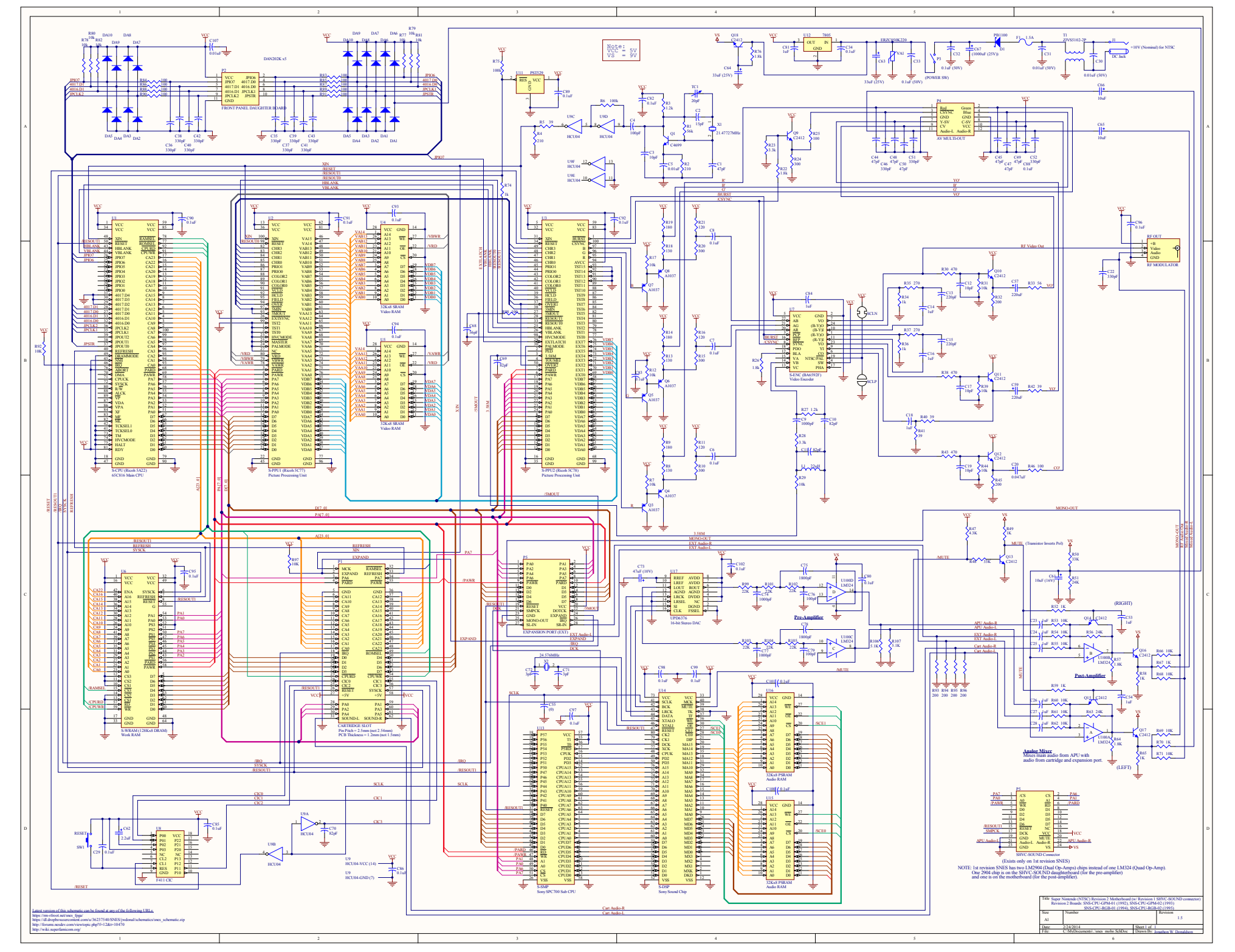

Note:
Vcc = 5V
Vs = 9V
S-CPU (Ricoh 5A22)
65C816 Main CPU
S-PPU1 (Ricoh 5C77)
Picture Processing Unit
S-PPU2 (Ricoh 5C78)
Picture Processing Unit
32Kx8 SRAM
Video RAM
S-ENC (BA6592F)
Video Encoder
S-WRAM (128Kx8 DRAM)
Work RAM
32Kx8 PSRAM
Audio RAM
S-SMP
Sony SPC700 Sub-CPU
S-DSP
Sony Sound Chip
EXPANSION PORT (EXT)
CARTRIDGE SLOT
Pin Pitch = 2.54mm (not 2.54mm)
PCB Thickness = 1.2mm (not 1.5mm)
16-bit Stereo DAC
FRONT PANEL DAUGHTER BOARD
AV MULTI-OUT
RF MODULATOR
SHVC-SOUND Connector
(Exists only on 1st revision SNES)
Pre-Amplifier
Post-Amplifier
Analog Mixer
Mixes main audio from APU with audio from cartridge and expansion port.
(RIGHT)
(LEFT)
(POWER SW)
DC Jack
+10V (Nominal) for NTSC
MUTE (Transistor Inverts Pol)
RF Video Out
3.58M
MONO-OUT
EXT Audio-R
EXT Audio-L
Cart Audio-R
Cart Audio-L
APU Audio-R
APU Audio-L
SCLK
SCL0
SCL1
/RESET
/RESOUT1
/SYSCK
/REFRESH
/RAMSEL
/CPURD
/CPUWR
/IRQ
/PAWR
/PARD
/SMOUT
/VBWR
/VRD
/VAWR
/VARD
/VDWR
/VDRD
/VAWR
/VARD
XIN
RESET
/RESOUT1
/RESOUT0
HBLANK
VBLANK
/BURST
/CSYNC
R
G
B
Y
C
CIC0
CIC1
CIC2
CIC3
PA7
PA6
PA5
PA4
PA3
PA2
PA1
PA0
PA[7..0]
PD[7..0]
DA[7..0]
A[23..0]
JPIO7
JPIO6
JPIO5
JPIO4
JPIO3
JPIO2
JPIO1
JPIO0
JPCLK2
JPCLK1
JPSTR
JPD0
JPD1
JPD2
JPD3
JPD4
JPD5
JPD6
JPD7
4017 D0
4017 D1
4017 D2
4017 D3
4017 D4
4017 D5
4017 D6
4017 D7
4016 D0
4016 D1
4016 D2
4016 D3
4016 D4
4016 D5
4016 D6
4016 D7
4016 CLK
4016 STR
Latest version of this schematic can be found at any of the following URLs:
https://rm-rfroot.net/snes_fpga/
https://dl.dropboxusercontent.com/u/36237540/SNES/jwdonal/schematics/snes_schematic.zip
http://forums.nesdev.com/viewtopic.php?f=12&t=10470
http://wiki.superfamicom.org
Title: Super Nintendo (NTSC) Revision 2 Motherboard (w/ Revision 1 SHVC-SOUND connector)
Revision 2 Boards: SNS-CPU-GPM-01 (1992), SNS-CPU-GPM-02 (1993)
SNS-CPU-BGIB-01 (1994), SNS-CPU-BGIB-02 (1995)
Size: A1
Number:
Revision: 1.5
Date: 2/24/2014
Sheet 1 of 1
File: C:\MyDocuments\...snes_mobo_SMDec
Drawn By: Jonathan W. Donaldson